

# rb\_vi\_bench

Reduced-basis dual cones for variational inequalities:  
algorithms, metrics, and measured results

Benchmark report

September 3, 2026

## Abstract

This report documents the benchmark assembled from three source repositories for reduced-basis (RB) model order reduction of contact problems. It states the problem setting, every selection algorithm under test, the exact computational definition of every metric, and the results measured on the three retained contact datasets. The object under study throughout is the *dual cone*: the reduced set from which Lagrange multipliers are drawn, on which non-negativity must hold. Sources: [BEE20] = Benaceur, Ern, Ehrlacher (ha1-02081485v2); [NDEE22] = Niakh, Drouet, Ehrlacher, Ern (ESAIM M2AN, 2022). Both papers number their algorithms and equations independently, so every citation below carries its paper tag.

## 1 Problem setting

The high-fidelity (HF) problem is a discrete variational inequality of contact type: find the displacement  $u(\mu) \in \mathbb{R}^N$  and multiplier  $\lambda(\mu) \in \mathbb{R}^m$  solving the saddle-point system

$$A u(\mu) + B(\mu)^\top \lambda(\mu) = f(\mu), \quad B(\mu) u(\mu) \leq g(\mu), \quad \lambda(\mu) \geq 0, \quad (1)$$

with complementarity between the last two. The sign condition  $\lambda \geq 0$  is physical: contact pressure pushes, it does not pull.

Reduction replaces  $\mathbb{R}^m$  by a low-dimensional set for  $\lambda$ . That set cannot be a *subspace*: a subspace closed under negation cannot enforce  $\lambda \geq 0$ . It must be a finitely generated **convex cone**

$$W_R^+ = \text{span}_+ \{\xi_1, \dots, \xi_R\} = \left\{ \sum_{r=1}^R c_r \xi_r : c_r \geq 0 \right\}, \quad \xi_r \geq 0. \quad (2)$$

With  $\xi_r \geq 0$  and  $c_r \geq 0$ , membership of the cone *implies* the sign condition. This is the single structural reason the whole family of methods exists, and it is why POD is inadmissible here: its modes have mixed signs ([BEE20] §5).

Two derived objects are used repeatedly. The **full snapshot cone**  $K_{\text{full}} = \text{span}_+ \{\theta_1, \dots, \theta_Q\}$  is everything the training multipliers generate; and the **non-negative orthant**  $W^+ = \{x \in \mathbb{R}^m : x \geq 0\}$  is the largest admissible set of all. Every method produces some  $W_R^+$  with  $W_R^+ \subseteq W^+$ , but—importantly—*not* necessarily  $W_R^+ \subseteq K_{\text{full}}$ .

The **cone projection**  $\Pi_K(x) = \arg \min_{y \in K} \|x - y\|$  is computed by non-negative least squares (NNLS) on the generator matrix. It is non-linear in  $x$ , which is what makes every error below more expensive than a linear projection and is the unit in which offline cost is counted.

## 2 Algorithms under test

All greedy methods are *hierarchical*: the cone at cardinality  $R$  is a sub-cone of the cone at  $R+1$ . This is verified rather than assumed (§4). NMF is not hierarchical, and that difference is visible in several metrics.

## 2.1 CPG — Cone-Projected Greedy ([BEE20], Algorithm 2)

Start from  $K_0 = \{0\}$ . At step  $r$ , select the training snapshot worst represented by the cone built so far, and append it unchanged:

$$q_r \in \arg \max_q \|\theta_q - \Pi_{K_{r-1}}(\theta_q)\|, \quad K_r = \text{span}_+\{\theta_{q_1}, \dots, \theta_{q_r}\}. \quad (3)$$

Because  $K_0 = \{0\}$  gives  $\Pi_{K_0} \equiv 0$ , the first step reduces to  $q_1 \in \arg \max_q \|\theta_q\|$ , which is [BEE20] Eq. (56). The generators are *selected snapshots*, so  $K_R \subseteq K_{\text{full}}$  holds by construction.

## 2.2 mCPG — modified CPG ([NDEE22], Algorithm 2)

Same selection loop, but the appended generator is a *residual* rather than the snapshot:  $\nu_r = (\theta_q - \Upsilon_r) / \|\cdot\|$  with  $\Upsilon_r \in K_{r-1} \cap (\theta_q - W^+)$ . The second constraint forces  $\nu_r \geq 0$ , i.e. membership of  $W^+$ , which is the property the method needs. It does *not* force membership of  $K_{\text{full}}$ : a difference of two elements of  $\text{span}_+\{\theta\}$  need not lie in  $\text{span}_+\{\theta\}$ . This is measured directly (§5).

## 2.3 ADG — Batch Normalized Angular-Defect Greedy

Not in either paper; contributed by the `greedy_algos` source. Two departures from CPG:

- **Angular selection.** It ranks candidates by the angle between a snapshot and its cone projection,  $\theta_K(x) = \angle(x, \Pi_K(x))$ , rather than by residual magnitude. On the normalized set  $S_{\text{norm}} = \{x/\|x\|\}$  the two induce the same ordering, since the residual is  $\sin \theta_K(x)$ ; un-normalized they differ. ADG is therefore *defined* on  $S_{\text{norm}}$ , and the tolerance becomes a genuine per-snapshot relative bound  $\varepsilon \in (0, 1)$  rather than one shared absolute threshold  $\varepsilon \max_q \|\theta_q\|$ .
- **Batch admission.** Every candidate attaining the maximal defect  $\theta_{\text{max}}$  enters in one round, not just one. This is what the method’s Theorem 3.5 certificate rests on.

Three ADG variants are benchmarked, each differing from the base in exactly one respect, so that any measured gap is attributable:

| key                       | varies                                                   | against          |
|---------------------------|----------------------------------------------------------|------------------|
| <code>adg</code>          | — (reference form)                                       |                  |
| <code>adg_momentum</code> | stopping rule: $ e(p) - e(p-1) /e(p-1) \leq \varepsilon$ | <code>adg</code> |
| <code>adg_k0</code>       | initialization: $K_0 = \{0\}$ , i.e. CPG’s first step    | <code>adg</code> |

The base ADG initializes not from  $K_0 = \{0\}$  but from the *pair* of snapshots at the largest mutual angle. This is its one departure from both papers, and `adg_k0` is its ablation: the first generator is instead  $\arg \max_q \|\theta_q\|$  taken on the *unnormalized* snapshots ([BEE20] Eq. (56)), normalized, after which the identical angular loop runs. Note that  $K_0 = \{0\}$  does not by itself define ADG’s first step: with the zero cone every defect is a right angle, all candidates tie, and batch admission would take the entire training set at once — Eq. (56) is the tie-break the papers supply. A consequence of the pair start is that base ADG has *no*  $R = 1$  state; its trajectory is  $R = 2, 3, 4, \dots$

## 2.4 Baselines

- **NMF** ([BEE20] §6.4): optimized non-negative atoms, free to sit anywhere in  $W^+$ . A genuine competitor, but cardinality-only — [BEE20] §5’s objection is precisely that “the user does not specify an error tolerance but only the cardinality” — and non-deterministic in its initialization.

- **Orthant** (reference, not a competitor): canonical directions  $e_i$  selected along mCPG’s own iteration, so the only difference from mCPG is the generator itself. Its pairwise angles are exactly  $90^\circ$ , the widest any cone in  $W^+$  can be. Its reported cost is the mCPG ordering run, not the (free) cone assembly.
- **POD** (*negative* control): the best unconstrained reconstruction at a given cardinality, and a direct measurement of the sign violation that motivates the cone construction. Its non-negativity check is *expected* to fail.

### 3 Metrics: how each is computed

#### 3.1 Precision

For a cone  $W_R^+$  with generator matrix  $\Xi$ , each snapshot is projected by NNLS and the residual is reported relative to a normalizer. Two normalizers are reported because they answer different questions:

$$\underbrace{\frac{\|\theta_q - \Pi(\theta_q)\|}{\max_{q'} \|\theta_{q'}\|}}_{\text{shared denominator}} \quad \text{and} \quad \underbrace{\frac{\|\theta_q - \Pi(\theta_q)\|}{\|\theta_q\|}}_{\text{per-snapshot}}. \quad (4)$$

The shared denominator keeps the number comparable against the tolerance that produced it; the per-snapshot form is what ADG’s tolerance actually bounds. They diverge whenever snapshot magnitudes spread. Train and test are reported separately.

**Non-negativity** is checked on the reconstruction:  $\text{nn\_max\_violation} = \max(0, -\min_i \hat{\lambda}_i)$ .

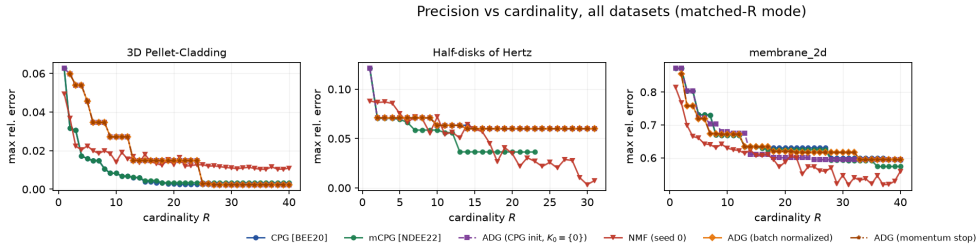


Figure 1: Test projection error against cardinality, all three datasets. Compressibility differs by orders of magnitude: Half-disks and Pellet-Cladding fall steeply, membrane barely moves — 125 snapshots on 497 dual dofs with a moving contact patch is simply not compressible to  $R \leq 40$ , which is a property of the problem and not of any method.

#### 3.2 Performance (offline cost)

Wall-clock is machine-dependent, so the primary cost is a *count of constrained solver calls* — NNLS, `lsq_linear` and `minimize` — captured by instrumenting the solvers during a fit. One caveat governs its reading: it counts *constrained* solves only, so NMF records exactly 0, meaning “issues none”, not “is free”. ADG likewise records 0 at  $R = 1, 2$ , because it seeds from a Gram-matrix argmin and NNLS first appears at  $R = 3$ .

#### 3.3 Stability

- **Gram conditioning:**  $\kappa(\hat{\Xi}^\top \hat{\Xi})$  on *column-normalized* generators. Normalizing is essential: without it the number is dominated by snapshot magnitude spread, and [BEE20] (raw generators) and [NDEE22] (normalized) would be incomparable. Undefined for  $R < 2$  and reported as such.

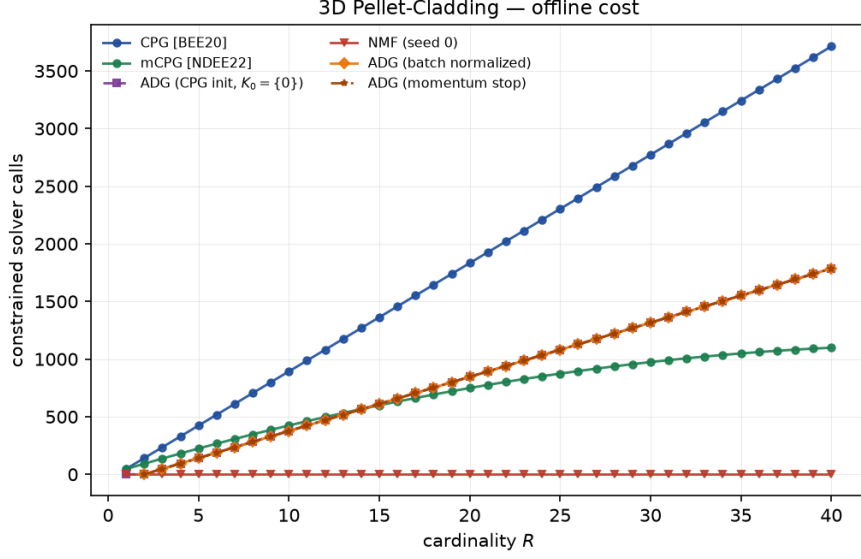


Figure 2: Offline cost on 3D Pellet-Cladding. CPG grows about twice as fast as ADG in constrained solves; mCPG plateaus once its residual stops finding new directions; NMF sits on the zero line throughout, which means it issues no *constrained* solves, not that it is free.

- **Orthogonality**  $e_{\text{orth}}$  ([NDEE22] Eq. 41):  $\|(I - \Pi_{K_r})(\nu_{r+1})\|$ , the sine of the angular defect of each accepted generator.
- **Determinism**: across-seed spread, which is the non-determinism [BEE20] §5 raises against NMF.

### 3.4 Cone geometry

This family scores the cone against  $K_{\text{full}}$  rather than against the finite snapshot set, by Monte-Carlo sampling of Dirichlet(1) conic combinations.

$$\text{missed (too small)} : \mathbb{E}_{x \sim K_{\text{full}}} \|x - \Pi_{W_R^+}(x)\|, \quad (5)$$

$$\text{excess (too large)} : \mathbb{E}_{x \sim W_R^+} \|x - \Pi_{K_{\text{full}}}(x)\|, \quad (6)$$

$$\text{two-sided} : \frac{1}{2}(\text{missed} + \text{excess}). \quad (7)$$

Neither direction implies the other: a cone can cover  $K_{\text{full}}$  perfectly while extending far beyond it, or sit strictly inside while missing most of it. The two-sided figure is the *average*, not the maximum. A maximum reports only whichever direction is larger and so silently changes which quantity it plots from one method to the next; the average is the same functional for every method, keeps both faults in view, and is still a metric on cones (symmetric, zero only when both vanish, triangle inequality inherited).

#### Extent (how much space the cone encloses)

Cut  $W_R^+$  by a hyperplane  $\langle x, u \rangle = h$  shared by every method, with  $u = (1, \dots, 1)/\sqrt{m}$  — strictly positive on  $W^+ \setminus \{0\}$ , so the cut is transversal for every admissible cone, and no coordinate is privileged. The section is the convex hull of  $p_i = h \xi_i / \langle \xi_i, u \rangle$ , and what is reported is its **mean width**, normalized by that of  $K_{\text{full}}$ ’s section:

$$\text{extent}(W_R^+) = \frac{w(S_R)}{w(S_{\text{full}})}, \quad w(S) = \mathbb{E}_{g \sim \mathcal{N}(0, I_m)} \left[ \max_i \langle p_i, g \rangle - \min_i \langle p_i, g \rangle \right]. \quad (8)$$

Three properties make this the right statistic, and each rules out an alternative that was tried and discarded:

- *Measured in the dimension of the space, not the number of generators.* The  $\xi_i$  are  $R$  rays in  $\mathbb{R}^m$  with  $R \ll m$ ; they are not a basis. Taking the section's  $(R-1)$ -volume builds the method's cardinality into the yardstick.
- *Monotone in  $R$  by construction.*  $W_R^+ \subseteq W_{R+1}^+$ , so the section can only grow; more vertices means a maximum over a superset, so the integrand rises pointwise in  $g$ . The Gaussian directions are drawn once from a fixed seed, so monotonicity holds per realization and Monte-Carlo noise cannot manufacture a decrease.
- *Positive on a thin cone*, where every ambient volume, solid angle or covered-sphere fraction is identically zero.

Values above 1 are meaningful: the cone is wider than  $K_{\text{full}}$ , which requires leaving it. Scale-invariance ( $\text{span}_+\{c\xi\} = \text{span}_+\{\xi\}$ ) means generator normalization cannot affect it.

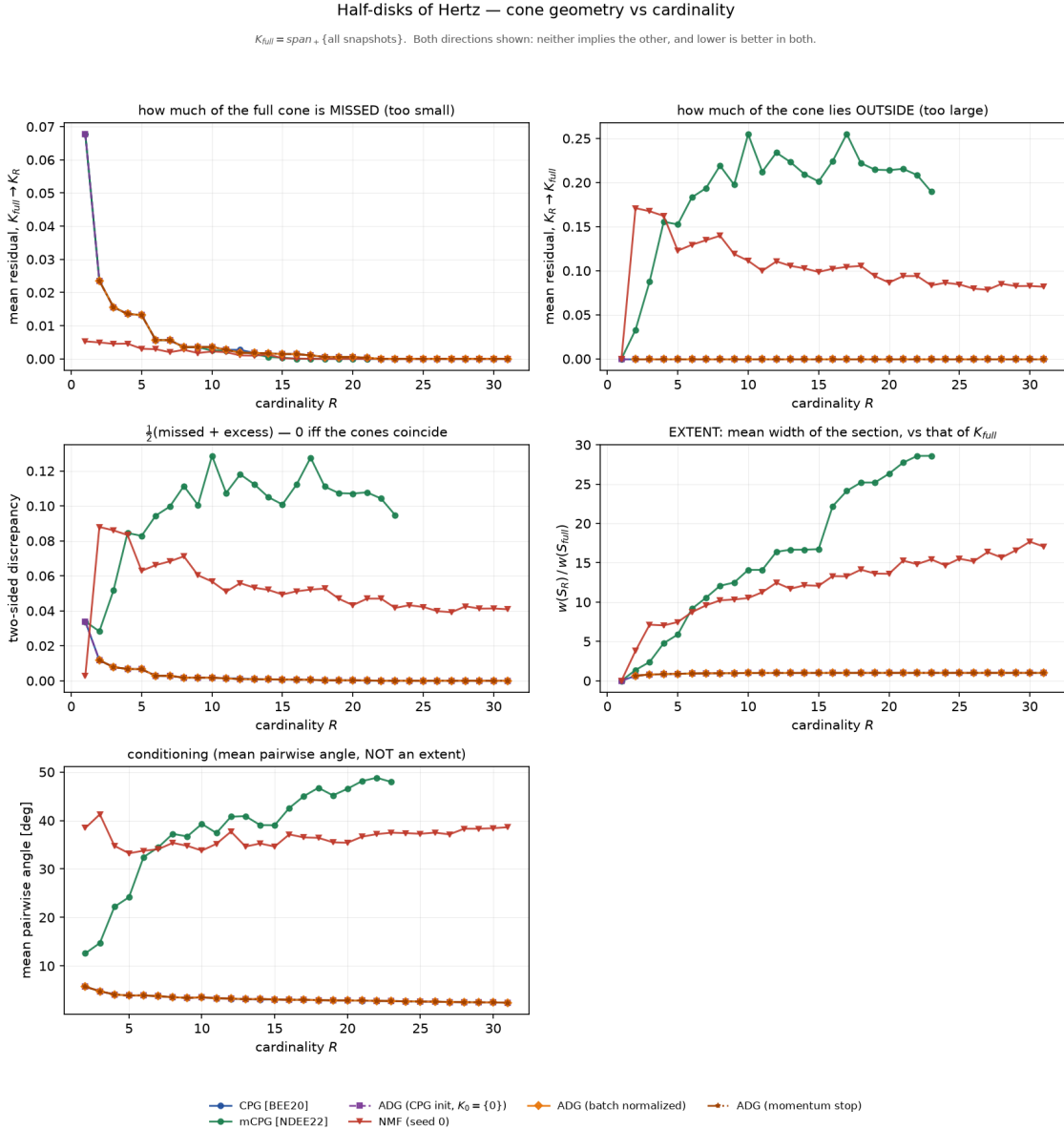


Figure 3: The five cone-geometry panels on Half-disks of Hertz. Read the two residual panels together: CPG and ADG sit flat at zero in *excess* because a cone spanned by snapshots cannot leave  $K_{\text{full}}$ , while mCPG lifts to 0.26. Extent and aperture disagree by design — aperture has mCPG rising smoothly while extent shows what the enclosed region actually does.

### Aperture (conditioning, *not* extent)

Mean pairwise angle between normalized generators. Retained as a conditioning diagnostic only. It is not a measure of enclosed space: it saturates once generators are mutually well separated, and a generator added strictly *inside* the cone enlarges the region not at all while still moving the mean.

### 3.5 Solved error (metrics.online)

Every metric above scores a cone against snapshots; none solves anything. This one assembles and solves the reduced saddle-point problem ([BEE20] Alg. 1 line 4) at each held-out parameter,

$$\hat{A} = V^\top A V, \quad \hat{B} = \Xi^\top B(\mu) V, \quad \min_{\alpha \geq 0} \frac{1}{2} \alpha^\top \hat{B} \hat{A}^{-1} \hat{B}^\top \alpha - \alpha^\top (\hat{B} \hat{A}^{-1} \hat{f} - \hat{g}), \quad (9)$$

and reports the relative error in *both* components against the HF solution. This is the *dual* form of the reduced problem, obtained by eliminating the primal coefficients in closed form; it is derived step by step in Appendix A, which also explains in what sense it is still a projection. The bound  $\alpha \geq 0$  is the discrete form of  $\text{span}_+$ , and it is why the recovered  $\hat{\lambda} = \Xi \alpha$  is non-negative exactly rather than approximately. Projection error and solved error are different functionals of the same cone: the solve minimizes an energy over  $W_R^+$ , it does not project the true multiplier onto it.

### 3.6 Cross-implementation agreement

Two independent transcriptions of CPG and of mCPG are retained deliberately. Agreement between them is checked at matched tolerance, so that a disagreement is a *finding* about an under-specified step rather than a rounding difference.

## 4 Experimental protocol

**Datasets.** Three contact problems on real geometry:

| key         | reported name       | dim $\times$ $n$ | geometry                     |
|-------------|---------------------|------------------|------------------------------|
| fem_lambda  | Half-disks of Hertz | $57 \times 50$   | mirrored contact line        |
| physics     | 3D Pellet-Cladding  | $7676 \times 94$ | $76 \times 101$ surface grid |
| membrane_2d | membrane (2-D)      | $497 \times 125$ | scattered 2-D patch          |

**Two modes, never mixed.** *Matched tolerance* gives every method the same  $\varepsilon$  and reports the  $R$  it needed; *matched cardinality* gives every method the same  $R$ . Only the second is used for comparison figures, because tolerances are not commensurable across methods (ADG’s is per-snapshot relative; CPG/mCPG’s is one shared absolute threshold).

**One greedy trajectory.** The cardinality sweep re-fits at each  $R$ . That is only legitimate if a greedy stopped at  $R$  yields exactly the first  $R$  generators of a longer run, so the property is asserted by test rather than assumed — and testing it found a real violation, since the upstream fixed-component helper answered  $R = 1$  for ADG through a different selection rule.

**Skips are recorded, never dropped.** A cell that does not run emits a `skip_reason`, because “not measured here” and “scored badly here” must never be confusable.

**Axes.** All figure axes are linear, with one documented exception: the Gram condition number, which is  $\geq 1$  by definition (so the zero-dropping failure of a log axis cannot occur) and spans eleven decades before the basis goes singular.

## 5 Results

All numbers below are medians over the matched-cardinality sweep  $R = 1, \dots, 40$  on the three datasets, from the current run (817 of 840 cells; 1848 s).

### 5.1 Non-negativity holds, exactly, for every cone method

Across all 817 cells and every cone method, the maximum non-negativity violation of the reconstruction is exactly 0. This is structural, not numerical:  $c \geq 0$  and  $\xi \geq 0$ . The POD negative control behaves as [BEE20] §5 predicts and fails: at  $R = 8$  it violates on 12.9% of entries on Half-disks and 89.4% on Pellet-Cladding. Driven through the reduced *solve*, POD’s recovered multiplier reaches  $-195$  where every cone method returns exactly 0.

### 5.2 Precision and cost

| dataset             | method | test err       | solves      | extent | two-sided      |
|---------------------|--------|----------------|-------------|--------|----------------|
| 3D Pellet-Cladding  | CPG    | <b>0.00261</b> | 1880        | 0.717  | <b>0.00024</b> |
|                     | mCPG   | 0.00339        | 762         | 5.74   | 0.0430         |
|                     | ADG    | 0.01482        | <b>893</b>  | 0.999  | 0.00232        |
|                     | NMF    | 0.01272        | 0           | 1.47   | 0.0489         |
| Half-disks of Hertz | CPG    | 0.0599         | 1240        | 0.999  | <b>0.00014</b> |
|                     | mCPG   | 0.0561         | 305         | 16.4   | 0.105          |
|                     | ADG    | 0.0599         | <b>589</b>  | 0.999  | 0.00016        |
|                     | NMF    | <b>0.0297</b>  | 0           | 14.4   | 0.0451         |
| membrane (2-D)      | CPG    | 0.630          | 4000        | 0.923  | 0.0839         |
|                     | mCPG   | 0.625          | 1849        | 0.947  | 0.0871         |
|                     | ADG    | 0.619          | <b>1900</b> | 0.906  | <b>0.0532</b>  |
|                     | NMF    | <b>0.591</b>   | 0           | 1.39   | 0.0891         |

**ADG costs consistently about half of CPG.** The median ratio of constrained solver calls, CPG over ADG at matched  $R$ , is  $2.15\times$ ,  $2.20\times$  and  $2.15\times$  on the three datasets — remarkably stable across problems differing by two orders of magnitude in dimension. On Half-disks ADG reaches the identical test error for that halved cost.

**NMF wins on raw reconstruction where it wins**, reproducing [BEE20]’s own honest finding, but at a cost the offline column cannot show and with cones that reach far outside  $K_{\text{full}}$  (extent 14.4 on Half-disks).

### 5.3 mCPG leaves the snapshot cone — and how much depends on the data

[NDEE22] Remark 4.3’s parenthetical claims  $\nu_r$  lies “in fact of  $\text{Span}^+(\{\theta_{q_n}\})$ ”. It does not hold as stated. The fraction of mCPG generators lying outside  $K_{\text{full}}$  (median over the sweep):

|                     | CPG   | mCPG         | ADG   |
|---------------------|-------|--------------|-------|
| 3D Pellet-Cladding  | 0.000 | 0.271        | 0.000 |
| Half-disks of Hertz | 0.000 | <b>0.917</b> | 0.000 |
| membrane (2-D)      | 0.000 | <b>0.951</b> | 0.000 |

What *does* hold is the  $W^+$  membership the same sentence claims first, and that is the part the construction rests on. CPG and ADG are at exactly 0 by construction, as they must be: their generators *are* snapshots. The extent metric confirms this independently — mCPG’s cone is  $5.7\times$  to  $16.4\times$  wider than everything the snapshots generate, which is only possible by leaving  $K_{\text{full}}$ .

### 5.4 ADG is the best conditioned of the greedy methods

Away from the rank cliff, ADG’s angular selection buys conditioning: on Pellet-Cladding it is  $45\times$  better conditioned than CPG at  $R = 5$  and  $125\times$  at  $R = 20$ . See §6 for the full study, which also shows that the *rate* of conditioning loss is set by the dataset (varying tenfold across the three) far more than by the method (well under twofold within any one).

## 5.5 ADG’s pair initialization: mostly irrelevant, and never decisive

Comparing base ADG against `adg_k0` (CPG’s first step) at matched  $R$ :

| dataset             | pair better | $K_0 = \{0\}$ better | identical |
|---------------------|-------------|----------------------|-----------|
| 3D Pellet-Cladding  | 0           | 0                    | 39 of 39  |
| Half-disks of Hertz | 0           | 0                    | 30 of 30  |
| membrane (2-D)      | 12          | <b>27</b>            | 0 of 39   |

On two of three datasets the two initializations produce *the same cone at every cardinality*, because the largest-magnitude snapshot already lies in the largest-angle pair. Where they differ, the CPG-style start wins more often. Offline cost is unaffected: the choice moves *which* snapshots are selected, never how many solves it takes. The pair initialization is therefore not earning its departure from the papers on this evidence.

## 5.6 Agreement between the duplicate transcriptions

CPG’s two independent transcriptions agree on **36 of 36** comparisons. mCPG’s do not: of 18 comparisons, 7 are exactly equivalent, 1 agrees within solver tolerance, 9 produce a *different cone of the same accuracy*, and 1 is genuinely divergent. This is a finding about [NDEE22] Algorithm 2 line 9, whose constrained shift is under-specified — the two implementations resolve it with different solvers, and the resulting cones differ.

## 6 Conditioning: a closer study

Conditioning deserves its own treatment because it is the quantity that decides whether a cone of a given cardinality is *usable*, independently of how accurate it is. The reduced solve inverts  $\hat{A}$  and forms  $\hat{B}\hat{A}^{-1}\hat{B}^\top$ ; recovering the coefficients  $c$  from a cone projection is an NNLS problem on  $\Xi$ . Both inherit  $\kappa(\hat{\Xi}^\top\hat{\Xi})$ . A cone that reaches a smaller projection error at a condition number of  $10^{17}$  has not necessarily bought anything.

### 6.1 What is measured, and on which convention

$\kappa$  is computed on **column-normalized** generators, and the choice is not cosmetic. [BEE20]’s CPG keeps raw selected snapshots; [NDEE22]’s mCPG normalizes. Measured on raw generators the number would partly report snapshot magnitude spread rather than cone geometry, and the two families would not be comparable. Both are recorded; their ratio shows how much the convention was worth:

| dataset             | method | $\kappa$ (normalized) | $\kappa$ (raw)        | raw/norm     |
|---------------------|--------|-----------------------|-----------------------|--------------|
| 3D Pellet-Cladding  | CPG    | $9.38 \times 10^6$    | $2.42 \times 10^6$    | 0.26         |
|                     | mCPG   | $1.12 \times 10^7$    | $1.12 \times 10^7$    | <b>1.000</b> |
|                     | ADG    | $8.15 \times 10^4$    | $1.29 \times 10^6$    | 15.8         |
| Half-disks of Hertz | CPG    | $1.38 \times 10^{17}$ | $5.67 \times 10^{16}$ | 0.41         |
|                     | mCPG   | $1.41 \times 10^5$    | $1.41 \times 10^5$    | <b>1.000</b> |
| membrane (2-D)      | CPG    | 104                   | 96.8                  | 0.93         |
|                     | mCPG   | 98.1                  | 98.1                  | <b>1.000</b> |

mCPG’s ratio is exactly 1.000 on all three datasets, which is the convention confirming itself: its generators are already unit vectors, so normalizing is the identity for it. For ADG on Pellet-Cladding the two differ by  $15.8\times$  — had the raw number been reported, ADG’s conditioning advantage there would have been hidden entirely.

membrane\_2d — ADG Initialization: largest-angle pair vs  $K_0 = \{0\}$   
 identical angular-defect rule, batch admission, normalization and stopping; only the first generator differs

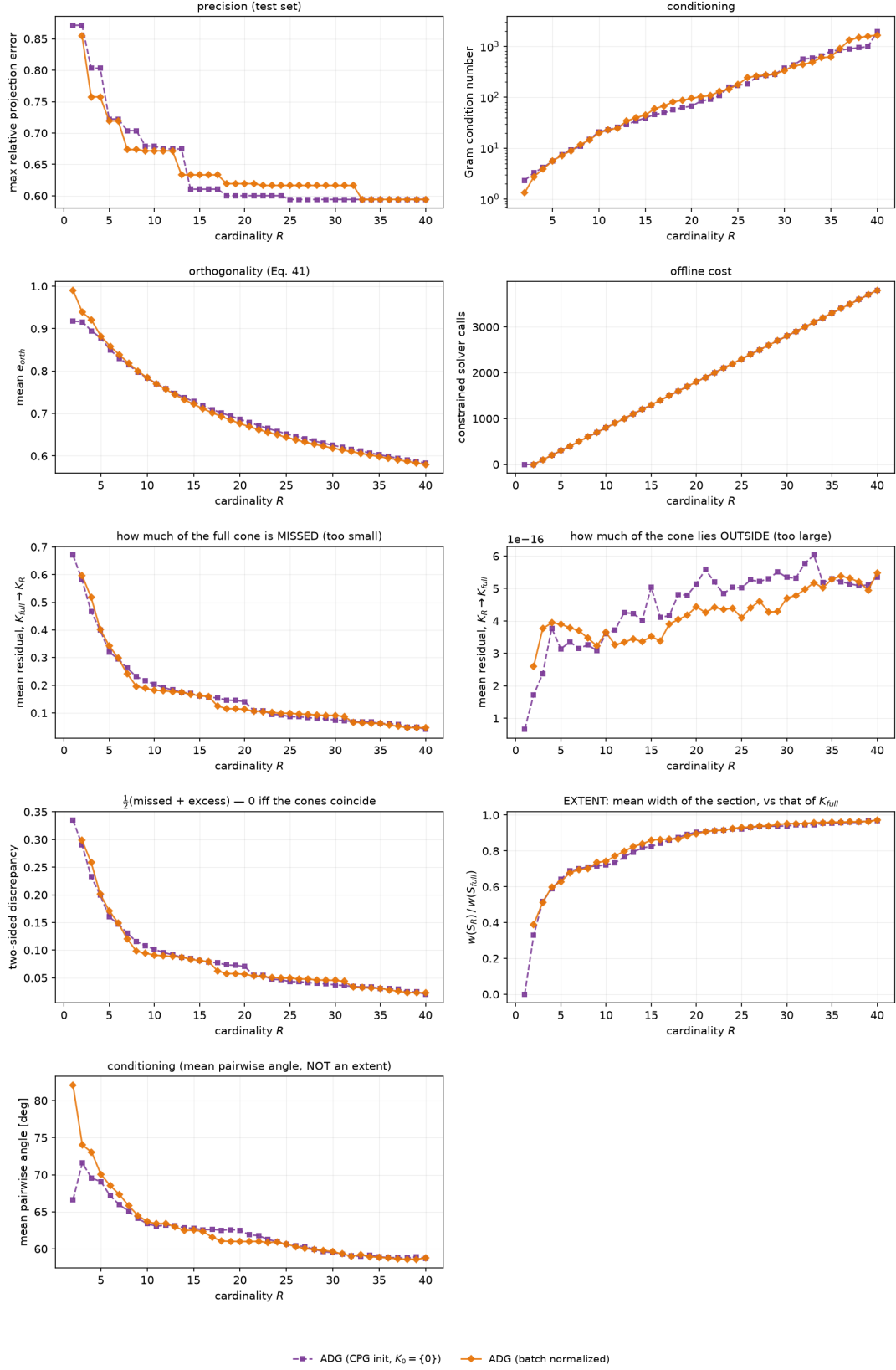


Figure 4: The initialization ablation on membrane — the one dataset of the three where the two arms differ at all. Every panel is the same algorithm apart from the first generator. On the other two datasets the curves coincide exactly at every cardinality.

## 6.2 Growth is geometric in $R$ , at a rate set by the dataset

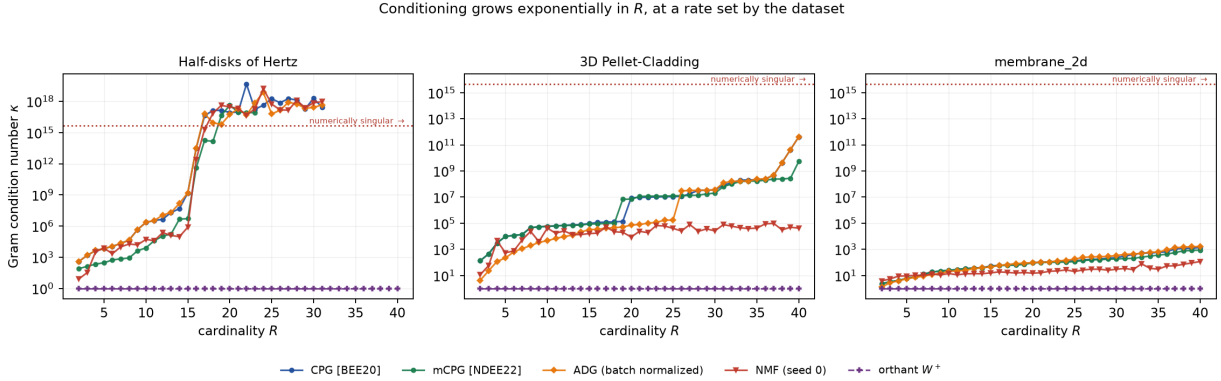


Figure 5:  $\kappa$  against cardinality. Note the log axis — this is the one panel in the benchmark that uses one, because  $\kappa \geq 1$  by definition so the zero-dropping failure of a log scale cannot occur, and the quantity spans eleven decades. The orphant sits at exactly 1 at every  $R$ : its generators are mutually orthogonal, so it is the best-conditioned admissible cone there is, and it bounds every curve from below.

On a log axis the curves are close to straight over most of their range, i.e.  $\kappa$  grows geometrically with each added generator — with one important exception, Half-disks, taken up in §6.3. Fitting  $\log_{10} \kappa$  against  $R$  over the non-singular range gives the rate in *decades gained per generator*:

| dataset             | CPG   | mCPG  | ADG   | NMF          |
|---------------------|-------|-------|-------|--------------|
| 3D Pellet-Cladding  | 0.180 | 0.157 | 0.222 | <b>0.051</b> |
| Half-disks of Hertz | 0.572 | 0.705 | 0.586 | 0.608        |
| membrane (2-D)      | 0.063 | 0.057 | 0.071 | <b>0.027</b> |

The rate is a property of the *dataset* far more than of the method: it varies by a factor of ten across datasets and by well under two across methods on any one of them. NMF is consistently the gentlest, which is what one expects from atoms that are optimized rather than selected — it is free to place a generator where it improves conditioning, whereas a greedy must take a snapshot that exists.

## 6.3 The collapse is the snapshot spectrum, not the algorithm

The Half-disks rates above are misleading taken alone, and the figure shows why: that dataset does not decay geometrically at all. It is flat to  $R \approx 15$  and then falls off a cliff, with every method collapsing together within two cardinalities of each other ( $R = 17$  for CPG and ADG, 18 for NMF, 19 for mCPG). A single fitted slope averages across that cliff and describes neither side of it.

The explanation is the numerical rank of the snapshot set:

| dataset             | $\dim \times n_{\text{train}}$ | $\sigma_2/\sigma_1$ | $\text{rank}_{10^{-8}}$ | conditioning behaviour               |
|---------------------|--------------------------------|---------------------|-------------------------|--------------------------------------|
| Half-disks of Hertz | $57 \times 31$                 | 0.026               | 17 of 31                | flat, then a cliff at $R \approx 16$ |
| 3D Pellet-Cladding  | $7676 \times 47$               | 0.057               | 41 of 47                | smooth staircase, never singular     |
| membrane (2-D)      | $497 \times 100$               | 0.380               | 100 of 100              | gentle, $\kappa < 10^4$ throughout   |

Once  $R$  exceeds the numerical rank, any further generator is a numerically dependent combination of those already selected, and no selection rule can avoid it: the information is not in the data.

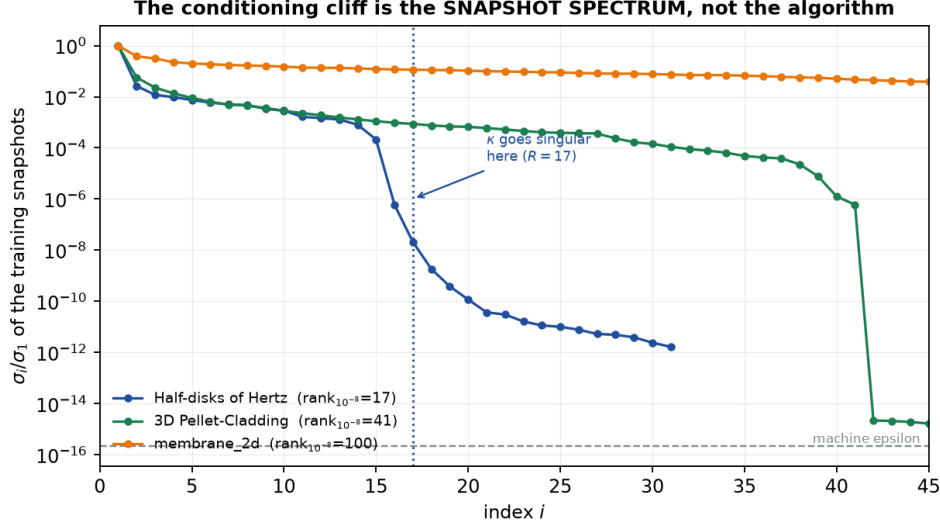


Figure 6: Normalized singular values of the training snapshots. The Half-disks spectrum drops from  $\sigma_{14} = 2.1 \times 10^{-4}$  to  $\sigma_{15} = 6.0 \times 10^{-7}$  to  $\sigma_{16} = 2.1 \times 10^{-8}$  — three decades in two indices. That cliff is at exactly the cardinality where every method’s conditioning explodes.

This is the single most useful conditioning result in the benchmark, because it is *predictive* — the spectrum is computable before any method runs, and it says where the usable range ends. On Half-disks that is  $R \approx 15$ , and reporting a Half-disks result at  $R = 30$  is reporting round-off.

## 6.4 Conditioning and aperture measure the same thing from two sides

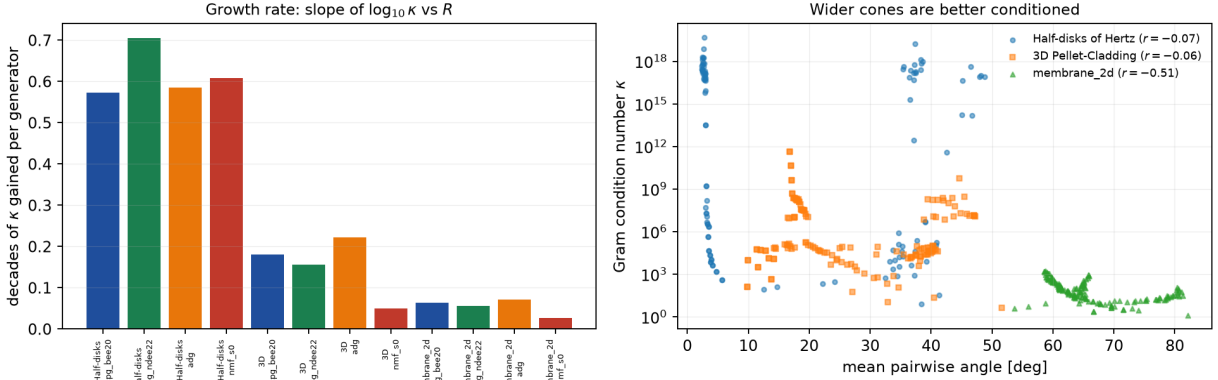


Figure 7: Left: growth rate per method and dataset. Right:  $\kappa$  against mean pairwise angle, pooled over methods and cardinalities.

The correlation between  $\log_{10} \kappa$  and mean pairwise angle is  $-0.70$ ,  $-0.59$  and  $-0.81$  on the three datasets: wider cones are better conditioned, consistently and strongly. This is why aperture is retained as a *conditioning* diagnostic in §3.4 and explicitly not as a measure of extent — it tracks  $\kappa$ , which is what [NDEE22] §4 is after when it asks for a well-conditioned Gram matrix, and it does *not* track how much space the cone encloses.

## 6.5 Which method to prefer, on conditioning grounds

Where the datasets can discriminate at all — that is, away from the rank cliff — ADG is markedly the best conditioned of the three greedy methods. On Pellet-Cladding at  $R = 5$  it reaches  $\kappa = 2.2 \times 10^2$  against CPG’s  $9.8 \times 10^3$ , a factor of 45; by  $R = 20$  the gap is  $7.1 \times 10^4$  against  $8.9 \times 10^6$ , a factor of 125. This is the expected consequence of its selection rule: choosing by *angular* defect rather than residual magnitude is, almost by definition, choosing generators that are far from the span of those already held. On Half-disks the cliff dominates and no method is distinguishable; on membrane every method sits near  $10^2$  and the differences do not matter.

## 7 Caveats

1. **No dataset currently drives a reduced solve.** The two 1-D obstacle toys were the only sources shipping  $A$ ,  $B(\mu)$ ,  $f$  and  $g$ ; with them removed, the inf-sup family and the solved-error metric report nothing on the three retained datasets. Both remain implemented and generic, and their tests build such a problem themselves.
2. **The orthant’s cost is mCPG’s ordering run**, not the cone assembly, which is free. Tracking mCPG is what buys the comparability; the price is that the cost column measures the tracking.
3. **Extent above 1 is not an error.** It means the cone is wider than  $K_{\text{full}}$ , which requires leaving it. mCPG and the orthant both do.
4. **The full-cone reference in the solved-error table is not a lower bound.** The reduced solve minimizes over the cone rather than projecting onto it, so a smaller cone can land closer.
5. **Half-disks results are meaningless past  $R \approx 15$** , and this is a statement about the data rather than the methods: the snapshot spectrum has numerical rank 17 of 31, so beyond that every generator is a numerically dependent combination of those already held (§6). The usable range is predictable from the spectrum before any method is run.
6. **Membrane errors are large in absolute terms ( $\approx 0.6$ )** for every method: 125 snapshots on 497 dual dofs with a moving contact patch is simply not compressible to  $R \leq 40$ . Comparisons there remain valid; the absolute level is a property of the dataset.

## Reproduction

```
python -m bench.runner --datasets fem_lambda physics membrane_2d \
    --subsample 200 --out results
python -m bench.runner --datasets fem_lambda physics membrane_2d --deltas \
    --cardinalities $(seq 1 40) --no-infsup --no-determinism \
    --subsample 200 --out results/sweep_dense
python -m bench.report --results results --out results/report.txt
python -m bench.figures --results results/sweep_dense --out results/figures --split
```

Main grid 174/210 cells in 919 s; dense sweep 817/840 cells in 1848 s; 145 tests passing.

## A Derivation of the reduced dual problem (Eq. 9)

### Step 0 — the HF problem is a constrained minimization

System (1) is the KKT system of

$$\min_u \frac{1}{2} u^\top A u - f^\top u \quad \text{subject to} \quad B u \leq g, \quad (10)$$

with  $\lambda \geq 0$  the multiplier of the inequality.  $A$  is symmetric positive definite (a stiffness matrix), which is what makes every elimination below legitimate.

### Step 1 — substitute the reduced ansatz

Take  $u \approx Va$  with  $a \in \mathbb{R}^N$  free in sign, and  $\lambda \approx \Xi\alpha$  with  $\alpha \geq 0$ . Testing the equilibrium equation against  $V$  and the constraint against  $\Xi$  gives the reduced KKT system

$$\hat{A}a + \hat{B}^\top \alpha = \hat{f}, \quad \hat{B}a \leq \hat{g}, \quad \alpha \geq 0, \quad (11)$$

with  $\hat{A} = V^\top AV$ ,  $\hat{B} = \Xi^\top B(\mu)V$ ,  $\hat{f} = V^\top f$ ,  $\hat{g} = \Xi^\top g$ . Equivalently: minimize  $\frac{1}{2}a^\top \hat{A}a - \hat{f}^\top a$  subject to  $\hat{B}a \leq \hat{g}$ .

### Step 2 — eliminate $a$

This is the step Eq. (9) compresses. The Lagrangian of the reduced problem is

$$L(a, \alpha) = \frac{1}{2}a^\top \hat{A}a - \hat{f}^\top a + \alpha^\top (\hat{B}a - \hat{g}). \quad (12)$$

$L$  is strictly convex in  $a$ , so the inner minimum is attained where  $\partial_a L = \hat{A}a - \hat{f} + \hat{B}^\top \alpha = 0$ , i.e.

$$a^*(\alpha) = \hat{A}^{-1}(\hat{f} - \hat{B}^\top \alpha). \quad (13)$$

Write  $s = \hat{f} - \hat{B}^\top \alpha$ , so  $a^* = \hat{A}^{-1}s$ . Substituting,

$$\begin{aligned} L(a^*, \alpha) &= \frac{1}{2}s^\top \hat{A}^{-1}s - \hat{f}^\top \hat{A}^{-1}s + (\hat{B}^\top \alpha)^\top \hat{A}^{-1}s - \alpha^\top \hat{g} \\ &= \frac{1}{2}s^\top \hat{A}^{-1}s - \underbrace{(\hat{f} - \hat{B}^\top \alpha)^\top}_{=s^\top} \hat{A}^{-1}s - \alpha^\top \hat{g} = -\frac{1}{2}s^\top \hat{A}^{-1}s - \alpha^\top \hat{g}. \end{aligned} \quad (14)$$

Expanding  $s^\top \hat{A}^{-1}s = \hat{f}^\top \hat{A}^{-1}\hat{f} - 2\alpha^\top \hat{B}\hat{A}^{-1}\hat{f} + \alpha^\top \hat{B}\hat{A}^{-1}\hat{B}^\top \alpha$  and *maximizing* the dual (equivalently minimizing its negative), the constant  $\frac{1}{2}\hat{f}^\top \hat{A}^{-1}\hat{f}$  drops and what remains is exactly Eq. (9):

$$\min_{\alpha \geq 0} \frac{1}{2} \alpha^\top Q \alpha - \alpha^\top c, \quad Q = \hat{B}\hat{A}^{-1}\hat{B}^\top, \quad c = \hat{B}\hat{A}^{-1}\hat{f} - \hat{g}. \quad (15)$$

The displacement is then recovered from (13), and  $\hat{\lambda} = \Xi\alpha$ .

### Step 3 — what $Q$ and $c$ mean

Neither is an arbitrary algebraic object.

- $Q = \hat{B}\hat{A}^{-1}\hat{B}^\top$  is the **Schur complement**, i.e. the compliance of the structure *seen from the contact interface*. Entry  $Q_{ij}$  is the interface displacement in generator direction  $i$  produced by a unit contact force in direction  $j$ . It is symmetric positive *semi*-definite.
- $c = \hat{B}\hat{A}^{-1}\hat{f} - \hat{g}$  is the **penetration of the contact-free solution**:  $\hat{A}^{-1}\hat{f}$  is the displacement with contact ignored,  $\hat{B}$  reads it at the interface, and  $\hat{g}$  is the gap. Positive entries are the penetration that contact forces must remove.

So Eq. (15) reads: *choose non-negative contact-force coefficients that undo the penetration  $c$ , weighted by the interface compliance  $Q$ .*

### Step 4 — in what sense it is a projection

The objection “this minimizes an energy rather than projecting” is half right. Completing the square in the  $\hat{A}^{-1}$  inner product  $\langle x, y \rangle_{\hat{A}^{-1}} = x^\top \hat{A}^{-1}y$ ,

$$\frac{1}{2}\alpha^\top Q\alpha - \alpha^\top c = \frac{1}{2}\|\hat{B}^\top \alpha - \hat{f}\|_{\hat{A}^{-1}}^2 + \alpha^\top \hat{g} - \frac{1}{2}\|\hat{f}\|_{\hat{A}^{-1}}^2, \quad (16)$$

verified numerically to  $2.8 \times 10^{-14}$  over random  $\alpha \geq 0$ . So the reduced solve *is* a projection — of the load  $\hat{f}$  onto the cone of attainable reactions  $\{\hat{B}^\top \alpha : \alpha \geq 0\}$ , in the **energy (compliance) metric**  $\hat{A}^{-1}$ , shifted by the linear gap term  $\alpha^\top \hat{g}$ . When  $\hat{g} = 0$  it is exactly that projection.

This is precisely why solved error and projection error are different functionals of the same cone, as §3.6 claims. The metric  $\Pi_K$  used everywhere else in this benchmark is the *Euclidean* projection of the *true multiplier* onto  $W_R^+$ . The solve performs an  $\hat{A}^{-1}$ -weighted projection of the *load*, offset by the gap. Two different objects, two different inner products: a cone can be good for one and poor for the other.

#### Step 5 — a caution about solving (15)

$Q$  is a Schur complement, so its rank is limited by the interface rather than by  $R$ : with more cone generators than the contact set can distinguish it is *singular*. On the synthetic obstacle problem used to test this metric,  $Q$  is rank-deficient in 60% of cells, one at rank 4 of 6 with  $\kappa(Q) = 3.7 \times 10^{16}$ .

That defeats a naive bound-constrained solver. L-BFGS-B reports **status=0** (converged) after eight iterations on such a cell while its KKT residual is still  $4 \times 10^{-3}$ , landing on a worse objective than the true optimum in 10% of cells. Because it reports success, nothing downstream notices. Writing (15) as a bounded least-squares problem via the eigen-decomposition  $Q = W\Lambda W^\top$  — where NNLS is an exact active-set method — reaches a KKT residual of  $1 \times 10^{-14}$  on the same cell. The implementation runs both and returns the lower objective, so it can never be worse than the reference it mirrors.